

DEMONSTRATION PROJECT — PoC INPUT DATA — NOT AN OFFICIAL PROJECT DOCUMENT

SHARED WATER RESOURCE DEVELOPMENT
PROJECT PLAN

Madhya Pradesh Shared Water Sustainability
Project
— Demonstration Project —

A fictional water-resource infrastructure scenario prepared solely as input data for the Sangam Proof-of-Concept (PoC) software evaluation.

Document Type	Shared Water Resource Development Project Plan
Project Type	Water Resource Infrastructure
State	Madhya Pradesh
Region	Central Madhya Pradesh
Estimated Project Budget	Rs. 250 crore
Expected Timeline	30 months
Document Status	PoC input data — fictional / non-official

This document does not represent, reference, or imply approval by any government authority, water resources department, or irrigation agency. All names, locations, figures, and ecological parameters are illustrative constructs created to exercise the Sangam decision-support software. No official ecological flow values, demand statistics, or government project identifiers are used.

1. Project Overview

The Madhya Pradesh Shared Water Sustainability Project (the "Project") is a fictional demonstration scenario representing a proposed water-resource infrastructure system in Central Madhya Pradesh. The scenario has been constructed to provide realistic, structured input data for testing the Sangam Proof-of-Concept (PoC) software, which evaluates infrastructure configurations against human demand, ecological, cost, and sustainability criteria.

Attribute	Detail
Project Name	Madhya Pradesh Shared Water Sustainability Project — Demonstration Project
Project Objective	Develop a water-resource infrastructure system that supports human development requirements while maintaining ecological sustainability and long-term resource availability
Estimated Project Budget	Rs. 250 crore
Expected Timeline	30 months

Proposed System Components

- Water storage infrastructure
- Distribution infrastructure
- Intake structure
- Pipeline / network
- Water management infrastructure
- Monitoring infrastructure

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2. Planned Human Requirement

The project is designed to support the categories of human requirement below. These are presented as planning assumptions for PoC modelling purposes and do not represent official demand statistics from any government or water-resources authority.

Requirement Category	Planning Assumption
Nearby Settlements	Assumed domestic water requirement from settlements in the vicinity of the proposed system
Agriculture / Service Requirements	Assumed water requirement to support local agriculture and service-sector activity
Future Development Demand	Assumed incremental demand growth associated with future development in the region

Note: Figures used elsewhere in this plan to represent these requirements are planning assumptions only, not verified demand studies.

3. Ecological Requirement

The planning process must consider the ecological factors below. Where a quantitative parameter would normally be required (e.g., an ecological flow value), it is explicitly marked as a PoC Simulation Parameter requiring field validation, rather than an invented official figure.

Ecological Factor	Planning Treatment
Seasonal Water Availability	Modelled as a variable input across seasons — PoC Simulation Parameter, requires field validation
Downstream Ecological Requirements	Downstream flow requirement is treated as a placeholder — PoC Simulation Parameter, requires field validation
Nearby Water Bodies	Assumed presence of water bodies near the proposed system, relevant to intake and storage siting
Environmental Sensitivity	Portions of the area are treated as environmentally sensitive for PoC modelling purposes
Seasonal Variability	Availability and demand are assumed to vary seasonally within the simulation
Resource Stress	Resource stress is modelled as a function of demand relative to assumed availability
Infrastructure Footprint	Physical footprint of storage, intake, and pipeline infrastructure is assumed as a planning input

4. Resource Constraints

Constraint	Description
Seasonal Availability	Water availability varies by season and constrains storage and supply planning
Human Demand	Settlement, agricultural, and future demand must be met within system capacity
Ecological Requirement	Downstream and habitat-linked ecological needs must be respected — subject to field validation
Infrastructure Cost	Estimated project budget is capped at Rs. 250 crore
Storage Capacity	Storage infrastructure must be sized to balance seasonal supply and demand
Distribution Efficiency	Network configuration should minimise losses and inefficiency in distribution
Long-Term Sustainability	The system should remain viable and maintainable over its operational life

5. Sustainability Objectives

#	Objective
1	Meet essential human requirements
2	Avoid unnecessary ecological pressure
3	Maintain long-term resource availability
4	Reduce resource-use inefficiency
5	Remain economically feasible
6	Support sustainable development

6. Alternative Configuration Requirement

The current proposal contains a preferred infrastructure location and configuration. However, if analysis indicates high ecological or resource stress, the project plan requires that Sangam evaluate alternative configurations. All alternatives must still satisfy the project's primary development objective of meeting human resource demand.

Alternative Configuration	Primary Adjustment Focus
Alternative Infrastructure Location	Relocates key infrastructure to reduce ecological or resource stress at the preferred site
Alternative Alignment	Reroutes pipeline/network alignment to reduce footprint or stress on sensitive areas
Alternative Capacity	Adjusts storage or distribution capacity to better balance demand and availability
Alternative Distribution Configuration	Modifies network layout to improve distribution efficiency
Balanced Resource Allocation Strategy	Seeks an even trade-off across human demand, ecological requirement, and cost

7. Evaluation Criteria Summary

Criterion	Indicative Evaluation Focus
Human Demand Fulfilment	Does the configuration meet assumed settlement, agricultural, and future demand?
Ecological Impact	Extent of pressure on downstream ecological requirements and sensitive areas (subject to field validation)
Resource Efficiency	Storage and distribution efficiency relative to seasonal availability
Cost	Capital cost relative to the Rs. 250 crore budget ceiling and lifecycle maintenance cost
Sustainability	Overall long-term balance between human, ecological, and resource outcomes

8. Project Planning Question

“How can the required human resource demand be met while maintaining ecological sustainability and long-term resource security?”

This question defines the core decision-support task for the Sangam PoC: to test whether the software can process the structured inputs above and produce a comparative evaluation of the preferred infrastructure configuration against the alternative location, alignment, capacity, distribution, and balanced-allocation configurations.

9. Document Status and Disclaimer

This entire document is a FICTIONAL DEMONSTRATION created solely to provide structured input data for testing the Sangam Proof-of-Concept (PoC) software. It does not describe an actual, planned, or approved government water-resource project. No real project names, government logos, official seals, official ecological flow values, demand statistics, government project identifiers, or claims of government approval are included or implied. Any resemblance to an actual project is coincidental and unintended. All figures (budget, timeline, demand, and ecological parameters) are illustrative placeholder values chosen for PoC testing convenience and are explicitly marked where field validation would be required.

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